Shared XYZZ affine recovery: report

Question and scope

Can the fixed `371-tau` evaluator recover affine coordinates for 128 independent scalar outputs with one shared inversion instead of one inversion per point? The candidate multiplies nonzero projective `ZZZ` values into prefix products, inverts the total once, and uses reverse products to recover each affine point. Table construction and fixture parsing precede the timed interval; point accumulation, prefix/reverse products, shared inversion and affine encoding are included. The 96 passes do not share an inversion across pass boundaries.

Qualified Tier 1 result

The frozen same-point panel completed 96 valid arms and all 48 pairs. Every output matched; source/binary/fixture artifacts were stable; all adjacent A/A ratios passed `[0.90, 1.11]`; observed cgroup throttling and execution-CPU steal increments were zero. The independent audit checks all arms, hashes, outputs and pair records and is reproducible from `analyze_batch_tier1.py`.

Metric, 12,288 outputs	Individual inversions	One shared inversion per 128 outputs
Mean online time	136.820 ms	61.706 ms
Mean time per scalar	11.134 µs	5.022 µs
Mean table preparation	3,954.631 ms	3,963.869 ms
Retained table bytes	26,035,988	26,035,988

The paired median individual/shared ratio is **2.209114×**, with repeat-bootstrap 95% interval [**2.1926, 2.2123**]. This qualifies the 128-output batch-throughput result for the observed RunPod Tier 1 instance. It does not measure isolated one-call latency; physical-host Tier 0 isolation is unverified. The parent README and raw archive record the complete panel, exact binary and source hashes, proof logs and receipt digests.

Correctness and proof

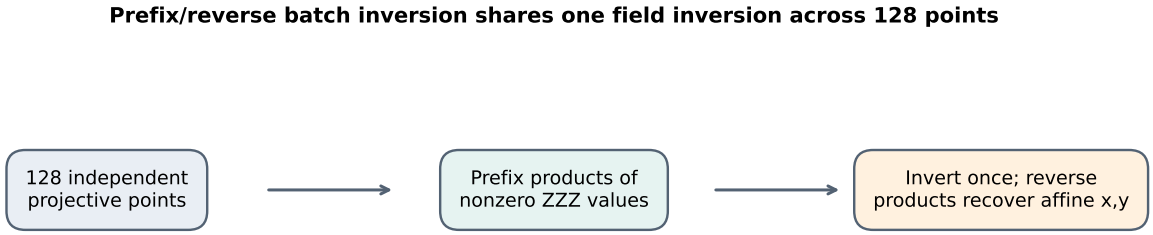
The hash-guarded overlay derives the XYZZ batch arm from the checked parent source. The native test checks the 128 frozen independent points against an independent binary multiplier and includes the identity boundary case. `BatchInverse.prefix_reverse_identity` is quantified over finite batch lengths and factor sequences with explicit prefix/reverse recurrences and final-inverse hypothesis; its checked proof and axiom audit are in `lean-check.log`. A Lean model of the concrete Montgomery-field bridge remains open.

The local audit command regenerated `tier1-batch-analysis.json` byte-for-byte. The figure script reads that committed audit and renders the throughput comparison plus the prefix/reverse dataflow diagram. This is an implementation-stage batch result; it does not report end-to-end scalar multiplication cost. The inspected parent scalar-pipeline record is unchanged by this report, so the stage graph is kept local rather than promoted into a full-pipeline dashboard.

Rebuild the figures and PDF with:

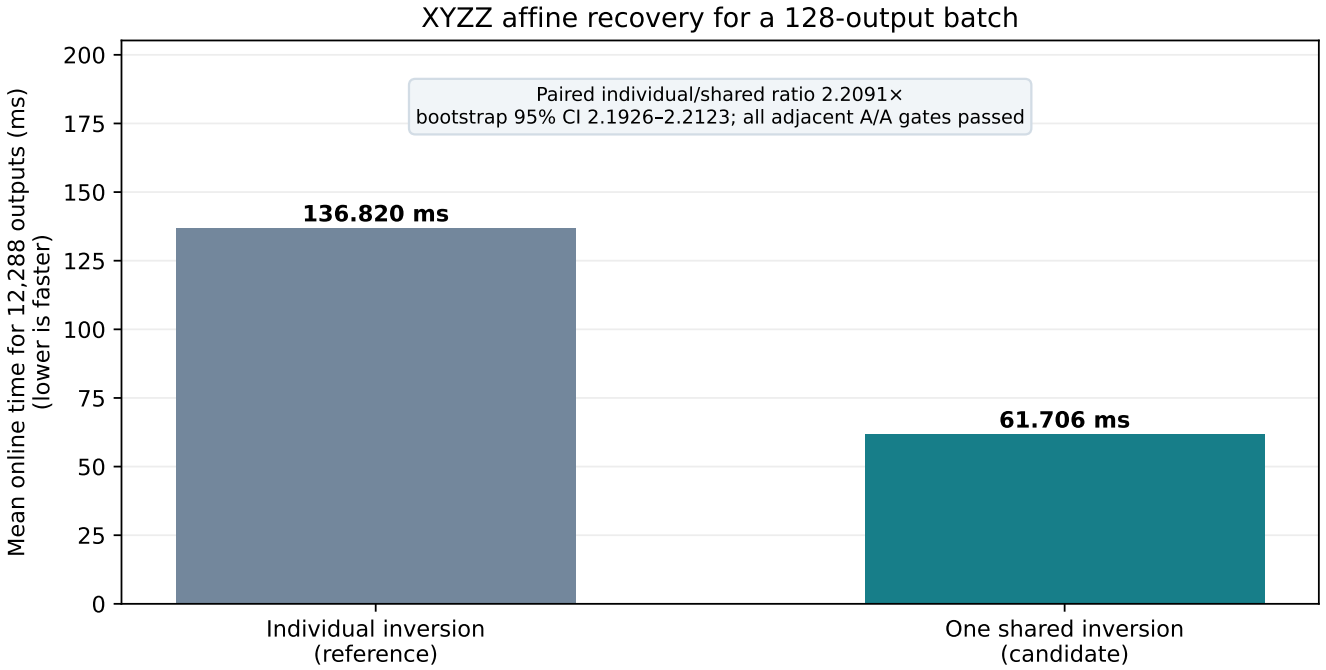
```
python3 experiments/prime-j0-xyzz-batch-inversion-20261015/visuals.py
pandoc experiments/prime-j0-xyzz-batch-inversion-20261015/REPORT-2026-10-19.md \
  --from=gfm --pdf-engine=xelatex -V geometry:margin=0.75in \
  -o experiments/prime-j0-xyzz-batch-inversion-20261015/REPORT-2026-10-19.pdf
```

Prefix/reverse diagram (SVG) · PDF



The batch consists of independent scalar outputs; no inversion is shared across the 96 passes.

Measured throughput figure (SVG) · PDF



CPU 13 · same 128 independent scalar outputs · table preparation excluded

Decision and next step

The measured batch method reduces online time for the specified 128-output workload. The next experiment varies chunk size under a separate predeclared panel; its qualified size-1 regression and size-2 gain are recorded independently in #911.